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* Formal Definition:

PDA = 6- tuple $(Q, \Sigma, \Gamma, \delta, q_0, F\}$

Q: start state

Σ: input Alphabet

Γ: stack Alphabet.

Ex: $L = \{ 0^n 1^n / n \geq 0 \}$ (re-visited)

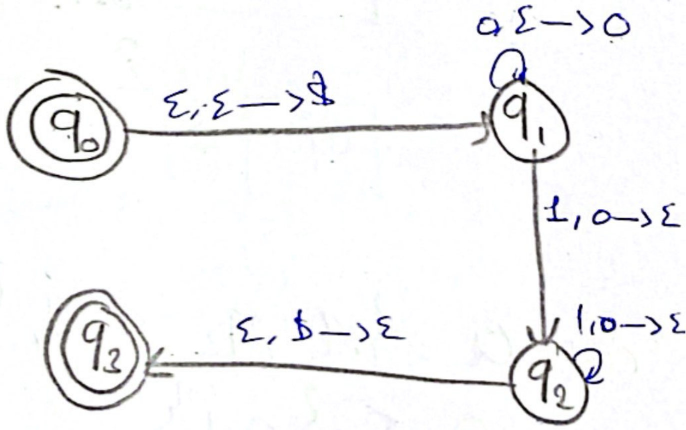
$M: = (Q, \Sigma, \Gamma, \delta, q_1, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Gamma = \{0, \$\}$

$\Sigma = \{0, 1\}$

$F = \{q_1, q_2\}$



δ:

input	0			1			ε		
what we are poping ⇐ stack	0	$	ε	0	$	ε	0	$	ε
q_0									$(q_1, \$)$
q_1			$(q_1, 0)$	(q_2, ε)					
q_2				(q_2, ε)				(q_3, ε)	
q_3									

=> CFL => CFG => conversion algorithm into PDA.

Algorithm: $Q = \{q_{start}, q_{loop}, q_{accep}\} \cup E$ / E = set of other states

ex:

CFG: $S \longrightarrow aTb \mid b$

$T \longrightarrow Ta \mid \varepsilon$